

ANUP RAJ NIROULA

Machine Learning Engineer (LLM Systems & MLOps) • Production ML • Scalable AI Systems

a.r.niroula@gmail.com | linkedin.com/in/anup-raj-niroula | github.com/ARNiroula | arniroula.github.io

EDUCATION

New York University

Master of Science in Computer Engineering – GPA: 4.0/4.0

New York, USA

September 2024 – May 2026

Kathmandu University

Bachelor of Engineering in Computer Engineering – GPA: 3.83/4.0

Bagmati, Nepal

August 2017 – March 2022

PUBLICATIONS

OpenMCP | EuroMLSys '26, ACM | Agustin Leon, **Anup Niroula**, Fraida Fund | pp. 456–465

[ACM Link](#)

- Built open-source benchmarking harness for MCP servers in agentic systems, evaluating 19 models across 6 infra configurations with full telemetry

EXPERIENCE

MedLaunch Concepts

Florida, USA

Machine Learning Engineer Intern (Team Lead)

January 2026 – May 2026

- Built CLARA, an LLM-powered healthcare compliance agent with a FastAPI backend, enabling automated policy analysis and workflow execution, reducing inference costs by 60% via prompt optimization and response caching
- Co-designed a hospital feasibility analysis system ingesting historical Centers for Medicare & Medicaid Services (CMS) data, enabling data-driven site selection decisions for clinical trial expansion

AI Engineer Intern

May 2025 – August 2025

- Built an event-driven Policy Analyzer using AWS Step Functions, Lambda, S3, and Bedrock to parallelize large-scale policy analysis workflows during accreditation cycles
- Engineered a serverless ETL pipeline with intelligent document routing (PDF, DOCX, images) using AWS Lambda and S3, processing 1,000+ concurrent documents with format-specific conversion flows
- Deployed a RAG-based chatbot with vector database retrieval, cutting query response time by 80% across 1,000+ queries

New York University

New York, USA

Graduate Teaching Assistant

January 2025 – December 2025

- Supported 100+ students in graduate-level Machine Learning courses, improving assignment performance by 20% through revised real-world problem sets

Nimble Clinical Research

New Jersey, USA

Software Engineer

October 2021 – May 2023

- Led a team of 6 engineers to build R-based clinical data pipelines and maintain Django–R microservice integrations
- Engineered a Report Automation System generating 100,000+ rows in under 60 seconds (9x performance increase)
- Devised an automated unit testing framework running test suites in CI/CD, ensuring code quality and early bug detection
- Optimized codebase, reducing server startup time in the Kubernetes cluster by 5x

PROJECTS

Spotify Buddies | MLflow, Ray, FastAPI, PostgreSQL, Redis, Grafana, Airflow, Docker

[GitHub Link](#)

- Built a production-scale recommendation system (KNN + Bayesian Personalized Ranking) serving 200K+ predictions/min with MLflow tracking and Ray Tune hyperparameter optimization
- Optimized the deployment pipeline, achieving 10x faster deployment (300s → 30s) while maintaining Docker image size under 2GB through multi-stage builds and layer caching
- Implemented real-time monitoring dashboards with Grafana and Prometheus, backed by PostgreSQL for feedback storage and system metrics tracking

News Classification (LoRA Fine-Tuning) | PyTorch, Transformers, Tensorboard

[GitHub Link](#)

- Fine-tuned RoBERTa and DeBERTa models using LoRA on news datasets with HuggingFace Transformers
- Applied domain-adaptive data augmentation by fine-tuning GPT-2 to generate 12k synthetic text samples from unlabeled data using custom randomized prompting techniques to enhance performance on target-specific news data
- Ranked 1st in an NYU Kaggle competition, achieving 93%+ OOD test accuracy

TECHNICAL SKILLS

Languages: Python, Go, SQL (Postgres, MySQL), JavaScript/TypeScript, R

ML/AI: PyTorch, Transformers, LLMs, LoRA/QLoRA, RAG, FAISS, LangChain, LangGraph

Backend: FastAPI, Django, Gin, Chi, Node.js

Data: Polars, Pandas, Tidyverse

MLOps: MLflow, Airflow, Ray, Docker, Prometheus, Grafana, Kubernetes, GitHub Actions, Argo CD, Langfuse

Cloud: AWS, GCP, Chameleon Cloud